Project Proposal

Project Title: Pet Ownership, Vaccination & Stray Control

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Introduction

The Pet Ownership, Vaccination & Stray Control system is a digital platform designed to manage pet registrations, vaccination tracking, and stray animal reporting. The system aims to address issues of low registration, rabies risk, and lack of centralized data by providing a single-window solution for pet owners, veterinarians, breeders, and municipal authorities.

Problem Statement

Currently, pet ownership and vaccination records are poorly maintained. This creates difficulties such as:

- Low registration rates leading to ineffective tracking.
- Increased risk of rabies due to missed vaccinations.
- Difficulty for authorities to monitor stray populations.

- Lack of proper communication between pet owners, vets, and municipalities.

The proposed system will address these challenges by enabling centralized, automated, and user-friendly tracking of pets and vaccination records, while supporting stray animal control.

Objectives

Primary Objectives:

- Provide secure pet registration with QR-based identification.
- Implement vaccination management with reminders for pet owners.
- Develop a module for reporting and managing stray animals.
- Enable veterinarians to update vaccination and health details.
- Provide dashboards for municipal authorities to track compliance.

Secondary Objectives:

- Support breeder licensing and compliance monitoring.
- Facilitate lost-and-found pet tracking.
- Provide a scalable solution for future integration with insurance and advanced analytics.
- Encourage responsible pet ownership and community safety.

Scope

The project will include the following core features:

Pet Registration:

- Secure owner-pet record creation with QR codes.
- Lost-and-found reporting.

Vaccination Management:

- Scheduler and reminders.
- Veterinarian access for updating records.

Stray Animal Reporting:

- Citizens can report stray sightings.
- Municipal dashboards for monitoring.

Breeder Licensing:

- Licensing requests and renewals.
- Compliance tracking.

Out of Scope:

- Direct treatment of animals.
- Payment/insurance services at this stage.

Requirements

Functional Requirements:

- Pet registration and QR ID generation.
- Vaccination scheduler and automated reminders.
- Veterinarian portal for vaccination updates.
- Stray reporting and tracking system.
- Lost-and-found pet reporting.
- Breeder licensing and compliance management.
- Municipal dashboards for monitoring.

Non-Functional Requirements:

- Secure authentication and encrypted data.
- User-friendly interface accessible via web/mobile.
- Scalable to support multiple regions.
- Reliable with backup and recovery options.
- Fast response times and real-time notifications.

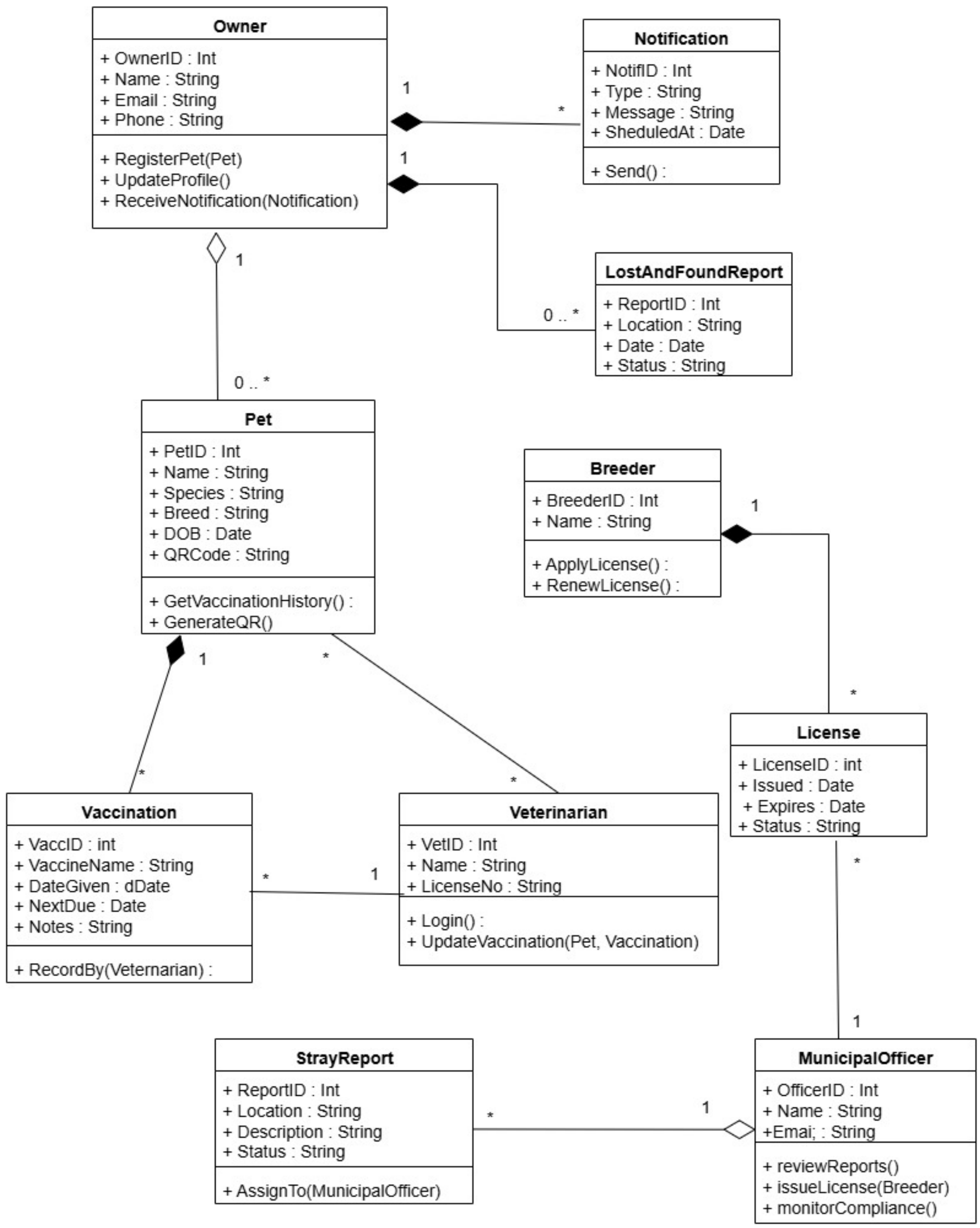
Methodology

The project will follow the Agile methodology of the Software Development Life Cycle (SDLC):

- Requirement Analysis: Gather requirements from client and stakeholders.
- Design: Create system architecture, database schema, and UI mockups.
- Development: Implement backend, frontend, and database functionalities.
- Testing: Conduct unit, integration, and acceptance testing.
- Deployment: Host on a cloud or municipal server.
- Maintenance: Address bugs, updates, and future improvements.

UML Class Diagram

Below is the UML class diagram representing the structure of the system:



Technologies

- Frontend: React JS (Web), Flutter (Mobile)
- Backend: Node.js
- Database: MongoDB
- Hosting: Cloud-based (AWS / Azure)
- Other Services: SMS/Email Gateway for vaccination reminders

Project Plan

Sprint 1: Setup of Pet Registration & QR ID system.

The first sprint will focus on setting up the core functionality of pet registration. During this stage, a secure module will be developed to capture essential details of pets and their owners, including species, breed, age, vaccination history, and owner contact information. A unique **QR-based identification system** will be integrated to generate IDs for each registered pet, ensuring authenticity and traceability. This sprint will also include creating an initial database schema and establishing the foundation for secure authentication. By the end of Sprint 1, the system should allow pet owners to successfully register their pets and obtain a unique QR ID that can be used for future tracking and vaccination updates.

Sprint 2: Development of Vaccination Scheduler and Reminder service.

In this sprint, the focus will shift toward ensuring timely vaccination and health record management. A **vaccination scheduler** will be developed, enabling veterinarians to log vaccine details and pet owners to receive timely notifications through SMS or email. The reminder service will be designed to notify owners in advance of upcoming vaccinations to reduce the risk of missing doses. This sprint will also include the integration of notification gateways and testing for real-time updates. By the end of Sprint 2, pet owners will be able to view vaccination schedules for their pets and receive automated reminders, while veterinarians can update vaccination records directly in the system.

Sprint 3: Implementation of Veterinarian Portal.

Sprint 3 will introduce a dedicated portal for veterinarians, enabling them to securely log in, manage patient records, and update vaccination or treatment details. The portal will provide veterinarians with a complete view of each pet's health and vaccination history, ensuring better decision-making. Role-based access control will be enforced so that veterinarians have appropriate permissions, while data privacy is maintained for owners. This sprint will also include initial testing of integration between the veterinarian portal and the main pet registry. By the end of Sprint 3, veterinarians will have a streamlined tool for maintaining accurate and up-to-date medical information for registered pets.

Sprint 4: Implementation of Lost-and-Found and Stray Reporting modules.

In Sprint 4, features will be implemented to assist both pet owners and local authorities in managing lost pets and stray animal populations. A **Lost-and-Found module** will allow owners to report missing pets and post details that can be viewed by others, improving the chances of reuniting lost pets with their families. In parallel, a **Stray Reporting module** will be developed to let citizens report stray animal sightings, which can then be tracked by municipal authorities for planning control measures. By the end of this sprint, the system will provide two-way communication between the public and authorities to address common animal welfare concerns.

Sprint 5: Development of Breeder Licensing and Admin Dashboards.

This sprint will focus on building modules for **breeder licensing** and **authority dashboards**. The breeder licensing system will allow breeders to apply for licenses, renewals, and maintain compliance records. Authorities will be able to review applications, approve or reject requests, and monitor breeder activities. The administrative dashboards will provide real-time insights into overall pet registration statistics, vaccination compliance rates, and stray reporting trends. By the end of Sprint 5, municipal authorities will have powerful monitoring tools, and breeders will have a structured way to ensure their operations comply with regulations.

Sprint 6: Testing, UI improvements, performance optimization, and deployment.

The final sprint will emphasize **system refinement and deployment**. Comprehensive testing will be conducted, including unit testing, integration testing, and user acceptance testing with sample stakeholders (pet owners, veterinarians, and municipal officers). User interface improvements will be made to ensure a clean, mobile-friendly, and intuitive design. Performance optimization will be carried out to guarantee fast data retrieval and reliable notifications. Backup and recovery mechanisms will also be implemented to improve system reliability. Finally, the system will be deployed on a cloud or municipal server, and training materials or documentation will be prepared for end-users. By the end of Sprint 6, the project will be ready for full-scale use with all core functionalities in place.